HridayMitra

A 90-second daily check-in that warns a family **3 days before** a heart-failure hospital visit. Offline. Voice-led. ₹4,330 to build.

72% caught · 0.06 false alarms / month

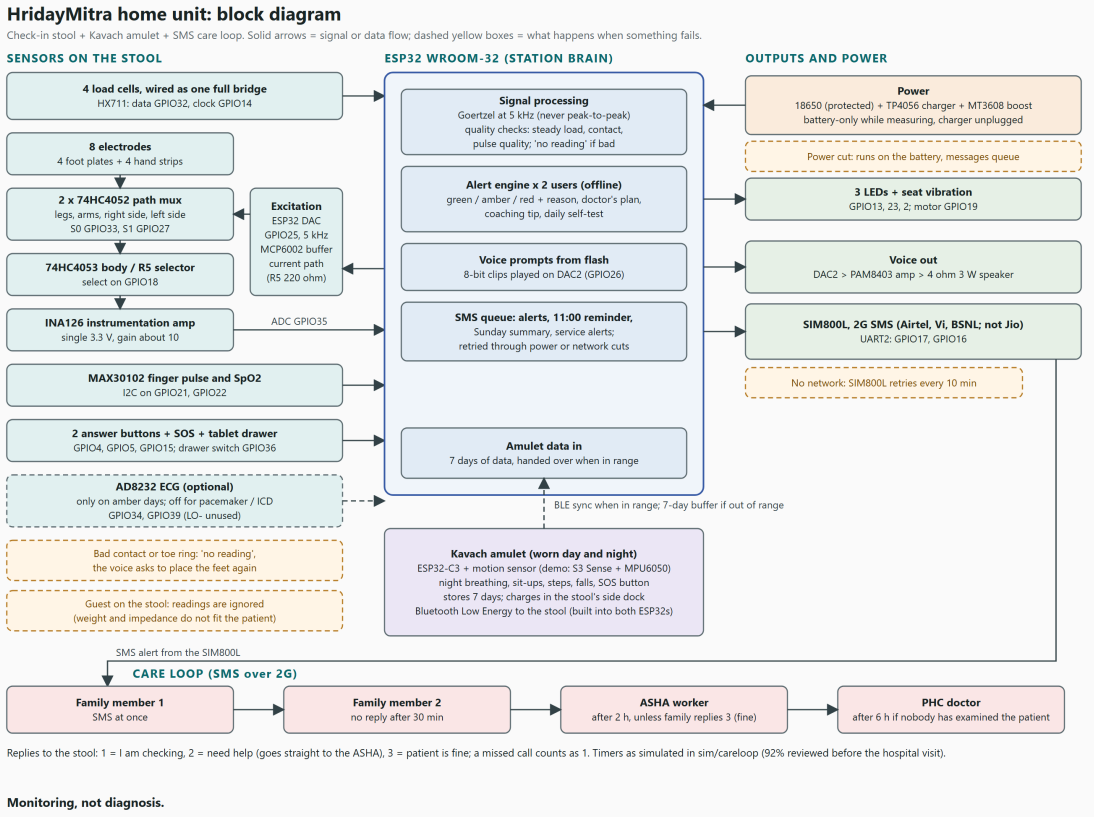
vs the weight-scale rule: 31% caught · 0.38 false alarms

The problem

- Heart failure fills the body with fluid **for days** before the emergency.
- Daily weighing (today's home standard) catches about **1 in 3** events and cries wolf.
- Elders will not wear sticky patches or vests; rural homes have power cuts, 2G and low literacy.
- An alert nobody acts on is worthless.

The solution

- **Check-in stool**: weight, 8-electrode bioimpedance (leg + chest fluid), pulse + SpO_2, stand-up speed, 3 spoken questions
- **Kavach amulet**: night breathing, sit-ups, activity, falls, SOS
- **Offline engine** + **SMS care loop**



More than a chair: a home care programme in a stool

- Speaks the **doctor's own sick-day plan**
- Coaching: salt, missed tablets, infection-like
- **Two people, one stool**: separate baselines
- Tablet drawer knows if medicine was taken
- Seat vibration for hard-of-hearing elders
- Daily **self-test** → "needs service" SMS
- 11:00 reminder, Sunday summary to the ASHA
- Village dashboard, OPD report, **ABDM/FHIR** export

How it decides: personal, multi-witness, explainable

- Learns **your** normal (median of days 7–35 ago)
- Scores only the worsening direction
- Red needs **two independent witnesses**: fluid · heart-lungs · daily life
- Every red says **why**, plus a cause tag (fluid vs infection vs missed pills)

SMS: HridayMitra RED for P0022.
Why: weight +1.9 kg; leg reading −7 %; night breathing +2/min.
Possible cause: fluid building.
Reply 1 checking, 2 need help, 3 fine.
Not a diagnosis; if breathless at rest call 108.

Results: frozen threshold, held-out patients

	HridayMitra	Weight rule	ML (logistic)
Sensitivity	**66%**	40%	80%
False alerts / patient-month	**0.06**	0.53	0.12
Specificity	98.7%	97.6%	-
Lead time	3 days (4 with amber)	2 days	2 days
1,000-patient external cohort	**72% / 0.06**	31% / 0.38	81% / 0.08

Fewest false alarms = families keep trusting the SMS.

We tried to break it

- **52/52** wrong-assumption worlds: still beats the weight rule
- Guest on the stool: rule fires 3.2 false alerts/month, we fire 0.10
- Half the days missing: still 47% caught
- Toe ring, wet feet, power cuts, electrode ageing, amulet left off: handled, tested
- Care loop: **92%** of alerts reach a clinician before the visit (one SMS: 27%)

Engineering proof

- LTspice: ≤ 0.05% error, contact-proof (tetrapolar, ratiometric)
- Monte Carlo of 100 cheap-part stools → found a flaw, **1 MΩ → 10 MΩ**, now 99% within 1.3%
- **Spectroscopy 5/20/50 kHz**: fluid outside cells +3.0%, foot position +0.0% (single frequency +3.0%)
- Patient safety: every single fault within IEC 60601-1 limits (≥ 12.8× margin)
- Self-calibrating reference: median error 0.16%, faults caught
- Mains hum: Goertzel 0.16% vs peak-to-peak 18%
- KiCad schematic from code, ERC clean, netlist auto-verified
- Engine in C++ on the ESP32: **100% identical** to Python over 3,231 days
- Whole firmware run through an emulated month: **26/26** checks, 3 bugs caught before flashing
- Real ECG, all 15 PhysioNet BIDMC CHF records: median heart-rate error 0.15 bpm (13/15 records within 0.5 bpm)

Cost and model

- Build cost **₹4,253–4,407** (limit ₹5,000)
- ≈ ₹2,598–3,031 at 100 units
- SMS ≈ ₹0.05 per patient-month

- Sold to PHCs / NGOs / insurers, placed by ASHAs
- One avoided admission pays for many stools
- No app, no smartphone, no internet

Honest limits and next steps

- Synthetic data: proves the method, not clinical accuracy
- Next: bench build vs simulation, ethics-approved pilot with lung-ultrasound comparison
- Record Hindi/Tamil prompts; tune SpO_2 against a reference oximeter
- Hybrid engine: rules + learned weights (+6–12 points, same alerts budget)

Ask: a pilot partner and seed support to take HridayMitra into 100 homes.